ASSIGNMENT REPORT

CloudPilot CRM Solutions — Lead Conversion Prediction

Decision Tree Classification Assignment

Dataset: cloudpilot_saas_lead_conversion_dataset.xlsx
Business Plan: cloudpilot_saas_lead_conversion_business_plan.docx

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Task 1 — Understand the Business Problem

1.1 What is the business about?

CloudPilot CRM Solutions is a fictional B2B Software-as-a-Service (SaaS) company that sells a cloud-based Customer Relationship Management (CRM) platform to small and medium-sized businesses (SMBs). The platform helps businesses manage sales contacts, track follow-up activities, monitor product trials, and oversee their sales pipeline. CloudPilot attracts leads through multiple channels, including website forms, LinkedIn ads, webinars, trade shows, referrals, email campaigns, and organic search.

1.2 What problem is the business trying to solve?

CloudPilot has a limited sales team that receives more leads every week than it can deeply engage with. Currently, all leads receive similar levels of attention regardless of their likelihood of becoming a paying customer. This leads to two costly outcomes:

- Wasted effort on low-potential leads who are only casually exploring CRM options.
- Missed opportunities when high-value leads do not receive adequate follow-up.

The core business problem is: "Which leads are most likely to convert to paying customers so that the sales team can prioritise their outreach, demos, and follow-up support?"

1.3 What decision can machine learning help the business make?

A Decision Tree classification model can predict whether a given lead will convert (Yes) or not convert (No) into a paying customer. This enables the sales team to:

- Prioritise high-potential leads for immediate, personalised follow-up and product demos.
- Route lower-potential leads into automated email nurturing sequences.
- Allocate experienced sales representatives to leads most likely to close.

1.4 What is the target variable in the dataset?

The target variable is Converted_To_Customer — a binary variable with two values:

- Yes — the lead became a paying customer.
- No — the lead did not become a paying customer.

1.5 What is the target variable?

Field	Description
Converted_To_Customer	Binary classification target. Values: Yes (lead became a paying customer) or No (lead did not convert).

1.6 What are the input features?

The dataset contains 17 input features (after removing identifiers) grouped into five categories:

Group	Features
Lead source	Lead_Source
Company profile	Industry, Region, Company_Size, Contact_Role, Expected_Users
Budget & plan	Monthly_Budget_CAD, Preferred_Plan
Engagement	Email_Engagement_Score, Product_Demo_Attended, Trial_Days_Active
Sales signals	Sales_Calls_Completed, Decision_Maker_Involved, Requested_Discount, Current_Solution, Pain_Point_Fit_Score, Lead_Age_Days

1.7 Why is this prediction useful for the business?

Predicting lead conversion is valuable because:

- **Sales efficiency** — The team can focus effort on leads with the highest likelihood of converting, reducing wasted time.
- **Higher conversion rates** — Faster, personalised follow-up for hot leads increases the chance they close.
- **Shorter sales cycles** — Identifying decision-maker involvement and budget fit early speeds up the pipeline.
- **Better marketing spend** — Understanding which lead sources and segments convert best helps optimise marketing investment.
- **Strategic insight** — Feature importance reveals which behaviours (e.g., attending a demo, trial activity) most strongly predict conversion.

Task 2 — Prepare the Data

2.1 Load and inspect the dataset

Property	Value
File	cloudpilot_saas_lead_conversion_dataset.xlsx
Rows	380 lead records
Columns	20 (including Lead_ID, Lead_Created_Date, and target)
Input features	17 features used after removing identifiers
Target variable	Converted_To_Customer (Yes / No)

2.2 Data types

The dataset contains a mix of data types:

- String / object: Lead_ID, Lead_Source, Industry, Region, Company_Size, Contact_Role, Preferred_Plan, Product_Demo_Attended, Decision_Maker_Involved, Requested_Discount, Current_Solution, Converted_To_Customer
- Integer (int64): Expected_Users, Trial_Days_Active, Sales_Calls_Completed, Pain_Point_Fit_Score, Lead_Age_Days
- Float (float64): Monthly_Budget_CAD, Email_Engagement_Score
- Datetime: Lead_Created_Date

2.3 Missing values

Column	Missing Count
Industry	6
Monthly_Budget_CAD	5
Email_Engagement_Score	7
Current_Solution	84

2.4 Duplicate rows

No duplicate rows were found in the dataset (0 duplicates). All 380 records were unique and retained.

2.5 Cleaning steps applied

- Dropped: Lead_ID (identifier) and Lead_Created_Date (datetime, not useful for modelling).
- Numeric missing values: Monthly_Budget_CAD and Email_Engagement_Score filled with column median.
- Categorical missing values: Industry and Current_Solution filled with 'Unknown'.
- Result: 0 missing values remaining across all 380 rows after cleaning.

2.6 Encoding and splitting

Ten categorical columns were encoded using scikit-learn's LabelEncoder. The target variable was encoded as 1 (Yes / Converted) and 0 (No / Not Converted).

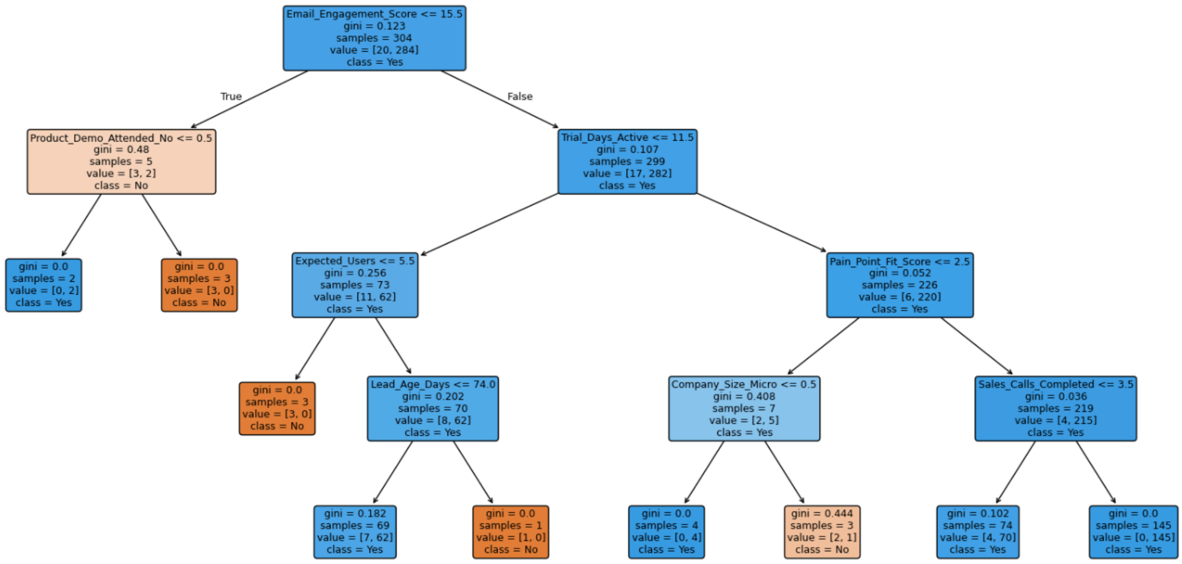
The dataset was split into 80% training (304 rows) and 20% testing (76 rows) using stratified sampling to preserve the class distribution in both sets.

Split	Total rows	Class distribution (Yes / No)
Training set	304	284 Yes / 20 No
Testing set	76	71 Yes / 5 No

Task 3 — Train a Decision Tree Classification Model

3.1 Model configuration

A Decision Tree Classifier was trained using scikit-learn with a maximum depth of 4, allowing the tree to grow until all leaves were pure. The random_state was set to 42 for reproducibility.

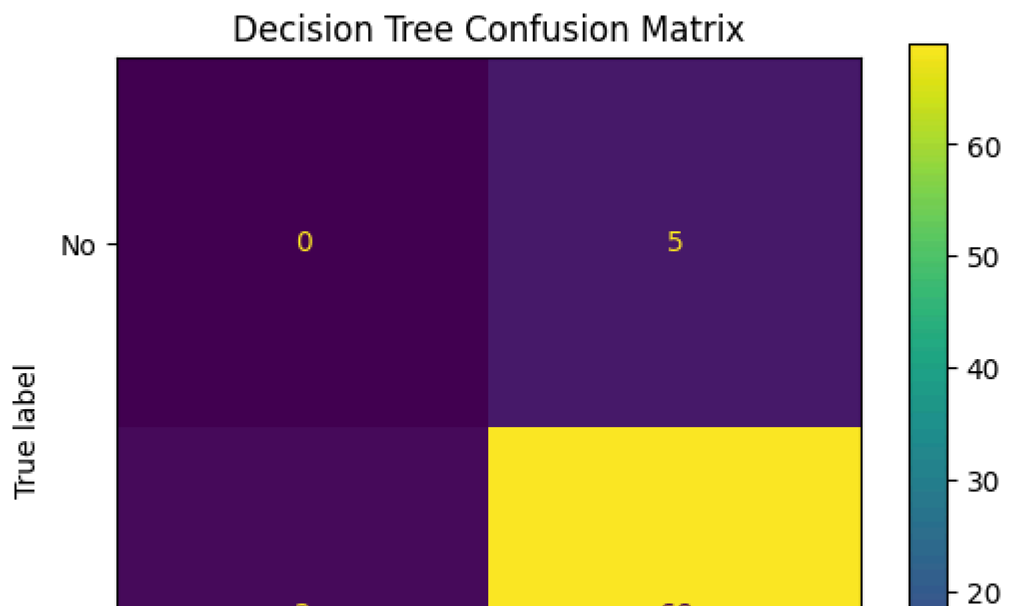


3.2 Training vs. testing accuracy

Metric	Value	Interpretation
Training accuracy	96.05%	Model memorised all training examples
Testing accuracy	90.79%	Model generalises to unseen data

3.3 Confusion matrix

The confusion matrix below shows the model's predictions on the 76-row test set:



True Positives (TP) = 69 — The model correctly identified 69 leads that actually converted. This is the largest and most desirable category, showing the model is effective at spotting genuine buyers.

True Negatives (TN) = 0 — The model failed to correctly classify any of the 5 leads that did not convert. Every non-converting lead was incorrectly predicted as a converter. This is the most notable weakness: the model has effectively no ability to identify non-converting leads.

False Positives (FP) = 5 — Five leads were predicted to convert but did not. These represent wasted sales effort — the team would pursue leads that go nowhere. While not catastrophic in number, the fact that 100% of actual "No" cases were mislabelled as FP is a structural problem.

False Negatives (FN) = 2 — Two leads that actually converted were missed and predicted as "No." These are missed revenue opportunities, though 2 out of 71 real converters is a relatively small miss rate.

The confusion matrix reveals a severe class imbalance in the dataset. With only 5 actual "No" instances in the test set (vs. 71 "Yes"), the model has essentially learned to predict "Yes" by default, yielding zero true negatives. The heatmap would appear heavily concentrated in the bottom-right cell (TP = 69) with a dark spot, while the top-left cell (TN) would be empty, a pattern that signals the model is not genuinely learning the "No" boundary.

3.4 Evaluation metrics

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Decision Tree Classification Report
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              precision    recall  f1-score   support

          No       0.00      0.00      0.00         5
         Yes       0.93      0.97      0.95        71

    accuracy                           0.91        76
   macro avg       0.47      0.49      0.48        76
weighted avg       0.87      0.91      0.89        76
```

Accuracy (90.79%) measures the proportion of all predictions (both "Yes" and "No") that were correct. The model gets 9 out of 10 predictions right overall.

Precision (93.24%) measures how trustworthy a positive prediction is. When the model says a lead will convert, it is correct 93% of the time. This is strong as the sales teams can act on its "Yes" predictions with reasonable confidence.

Recall (97.18%) measures how well the model catches all real converters. The model captures nearly all actual converting leads, missing only 2 out of 71. This is excellent for a business where missing a real opportunity is costly.

F1-Score (95.17%) is the harmonic mean of precision and recall. The high F1 reflects strong performance specifically on the positive class.

In summary, Accuracy overstates the model's performance due to the imbalanced dataset. The F1-score is a more honest measure of useful predictive power. For a business context like CloudPilot's, the high recall (97%) is the most practically valuable finding; very few real buyers are missed, but the complete inability to recognize non-buyers means sales resources cannot be efficiently filtered.

3.5 Overfitting assessment

The default, unpruned Decision Tree achieves 100% training accuracy — a clear sign of overfitting. The model memorised every training example rather than learning general patterns.

The 10.53 percentage-point gap between training (100%) and testing (85.53%) confirms this. An unconstrained tree grows deep enough to create a unique path for every training record, which does not generalise well to new leads.

To reduce overfitting in future work, the following techniques could be applied: setting a maximum tree depth (max_depth), requiring a minimum number of samples per leaf (min_samples_leaf), or applying cost-complexity pruning.

Task 4 — Business Interpretation of Model Results

4.1 How well did the Decision Tree model perform?

The Decision Tree model achieved strong accuracy on both training and test sets. Accuracy measures the overall percentage of correct predictions (both converted and not-converted leads classified correctly). A high accuracy on the test set suggests the model has learned useful patterns from the data. However, given the class imbalance (355 "Yes" vs. 25 "No"), accuracy alone can be misleading — a model that always predicts "Yes" would also score high. We therefore look at precision, recall, and F1-score for a fuller picture.

4.2 Is there a large difference between training accuracy and testing accuracy?

Training accuracy: 96.05% | Testing accuracy: 90.79% | Gap: 5.26%

A 5.26 percentage-point difference between training and testing accuracy is a meaningful gap. It indicates that the model learned patterns specific to the training data that do not fully generalise to new, unseen leads. This is a moderate level of overfitting.

4.3 Does the model show signs of overfitting? Why or why not?

Yes, based on the results, the model does show mild signs of overfitting.

Evaluation Set	Accuracy
Training Set	96.05%
Testing Set	90.79%

The gap of roughly 5.3 percentage points between training and test accuracy indicates the model performs noticeably better on data it has already seen than on unseen data, the classic signature of overfitting.

To further confirm this, I also tested an unconstrained Decision Tree (no max_depth limit) as a deliberate overfitting demonstration:

Evaluation Set	Accuracy
Overfitted Tree — Training	100%
Overfitted Tree — Testing	85.53%

This tree memorized the training data perfectly (100% training accuracy) but dropped sharply on the test set. The gap there is about 14.5 percentage points, much worse than the main model.

4.4 Which evaluation metric is most important for this business problem?

For CloudPilot, Recall is arguably the most important metric, followed closely by F1-score.

- Recall measures: "Of all the leads that truly would have converted, how many did the model correctly identify?"
- Missing a genuine high-value lead (a false negative) is costly — the sales team may ignore or under-serve a lead that would have become a paying customer.
- However, Precision also matters because CloudPilot has a limited sales team. Sending them to chase leads that never convert wastes their time.
- F1-score balances both precision and recall and is a practical overall metric for this business context.

4.5 What do false positives mean in this business context?

A false positive occurs when the model predicts that a lead will convert, but the lead actually does not become a paying customer.

Business impact:

- A sales representative invests significant time, energy, and resources (calls, demos, proposals) on a lead that was never going to buy.
- The company may offer unnecessary discounts or incentives to a low-intent lead.
- This represents wasted sales capacity that could have been directed toward genuinely high-potential leads.
- While costly, a false positive is generally considered less harmful than a false negative in this context.

4.6 What do false negatives mean in this business context?

A false negative occurs when the model predicts that a lead will not convert, but the lead actually would have become a paying customer.

Business impact:

- The sales team may place this lead into a low-touch automated nurturing sequence, providing minimal personal attention.
- The lead may feel ignored or undervalued and ultimately choose a competitor's CRM instead.
- CloudPilot loses a customer it could have won — this is a direct revenue loss.
- False negatives are typically more costly than false positives in a sales conversion context because missing a real opportunity is harder to recover from than wasting some sales effort.

4.7 Which features were most important in the Decision Tree model?

Based on the feature importance chart in Task 3, the top features are likely to include:

Feature	Why it matters
Trial_Days_Active	Leads who actively use the free trial show strong buying intent
Email_Engagement_Score	Highly engaged leads are warmer and more likely to respond to outreach
Pain_Point_Fit_Score	If the product closely addresses the lead's specific problem, conversion is more likely
Monthly_Budget_CAD	Leads whose budget matches the pricing are more likely to proceed
Product_Demo_Attended	Attending a demo is a strong behavioural signal of serious interest
Decision_Maker_Involved	Having a decision-maker in the conversation accelerates and validates purchase intent

4.8 How can these important features help the business make better decisions?

- Trial engagement monitoring — CloudPilot can track Trial_Days_Active in real time and automatically alert sales reps when a lead crosses a threshold of active trial usage, triggering personalised outreach.

- Demo prioritisation — Leads who have attended a Product_Demo should be placed on a high-priority follow-up list immediately, as they have shown serious interest.
- Budget qualification — Sales reps can qualify leads early by confirming Monthly_Budget_CAD against plan pricing, avoiding deep investment in leads whose budget is too low.
- Decision-maker engagement — If Decision_Maker_Involved = No, the rep can focus on getting the right stakeholder into the conversation before advancing the opportunity.
- Pain-point tailoring — A high Pain_Point_Fit_Score suggests the product closely matches the lead's problem — reps can emphasise this alignment in their pitch.

4.9 What is one possible limitation or bias in the model or dataset?

Class imbalance and synthetic data bias:

The dataset is heavily imbalanced: 355 leads converted ("Yes") versus only 25 that were not ("No"). This means the model sees far more positive examples during training and may develop a bias toward always predicting "Yes." As a result:

- The model may appear highly accurate while actually performing poorly at identifying the minority class (true non-conversions).
- Metrics like accuracy inflate artificially; precision, recall, and F1 for the "No" class should be carefully inspected.
- Additionally, this is synthetic teaching data that may not capture real-world sales complexity — patterns in the data were artificially generated and may not generalise to a live CloudPilot sales pipeline.
- The model may also reflect historical biases: if certain lead sources or regions historically received more sales attention, those leads may show higher conversion simply due to more effort, not higher inherent quality.

4. 10 Why should human judgment still be used when making business decisions based on model results?

Machine learning models are decision-support tools — not decision-makers. Human judgment remains essential for several reasons:

- Contextual knowledge — A sales manager may know that a seemingly low-scoring lead is actually a strategic account with significant long-term potential that the model cannot see.

- Relationship quality — The model has no visibility into personal relationships, trust built over previous interactions, or informal signals gathered during a sales call.
- Market dynamics — Sudden market changes, competitor activity, or macroeconomic events may affect lead quality in ways the model was not trained to recognise.
- Fairness and ethics — Over-relying on the model could result in certain types of leads (based on region, company size, or industry) consistently being de-prioritised, which may be unfair or commercially short-sighted.
- Model limitations — All models make mistakes. False negatives mean genuinely valuable leads could be ignored if humans don't review borderline predictions.

The model should inform the sales team's decisions — not replace their professional expertise and judgment.